of the software was supposed to use the SI system of units for all calculations. One group of software engineers used the Imperial system of units. This mismatch only caused a problem at the stage when the calculations concerned with achieving orbit around Mars were executed. This time the loss to the public purse was 125 million dollars.

These examples illustrate the public interest in successful software engineering. There is a strong argument that the correct application of the code of ethics with respect to specification, development and testing of software could have saved a lot of money.

A different type of disaster is the system that never gets built. In 2011 the UK government scrapped the National Programme for IT in the NHS (National Health Service), which had been commissioned in 2002. The project failed to produce a workable system. The estimated amount spent on the program was 12 billion pounds. The initial estimated cost was less than three billion pounds. In examples like this the software engineers are not to blame, but if correctly applied, the part of the code of ethics specifically targeted at project management would not have allowed this type of failure to occur.

In the three examples outlined above, the public concern was solely related to the costs associated with a failed project. There was no public concern relating to the ethics of the endeavour itself. In contrast there are many areas associated with computer-based systems where there is public concern about the nature of the endeavour or at least about what it has led to. Here are some examples:

- powerful commercial companies being able to exert pressure on less powerful companies to ensure that the powerful company's products are used when alternatives might be more suitable or less costly
- companies providing systems that do not guarantee security against unauthorised access
- organisations that try to conceal information about a security breach that has occurred in their systems
- private data transmitted by individuals to other individuals being stored and made available to security services
- social media sites allowing abusive or illegal content to be transmitted
- search engines providing search results with no concern about the quality of the content.

There is by no means a consistent public attitude to concerns like this. This makes it difficult for an individual software engineer to make a judgement with respect to public good. Even if the judgement is that a company is not acting in the public good, it will always be difficult for an individual to exert any influence. There are recent examples where individuals have taken action which has resulted in their life being severely affected.

Discussion Point:

This section has deliberately been presented in generalisations. You should carry out a search for some individual examples and then consider actions that could have been taken and justified as being for the public good.

10.04 Ownership and copyright

Copyright is a formal recognition of ownership. If an individual creates and publishes some work that has an element of originality, the individual becomes the owner and can therefore claim copyright. An exception is if the individual is working for an organisation. An organisation can claim copyright for a published work if it is created by one or more individuals that work for the organisation. Copyright cannot apply to an idea and it cannot be claimed on any part of a published work that was previously published by a different individual or organisation.

KEY TERM

Copyright: a formal recognition of ownership of a created and published work

Copyright can apply to any of:

- a literary (written) work
- a musical composition
- a film
- a music recording
- a radio or TV broadcast
- a work of art
- a computer program.

The justification for the existence of copyright has two components. The first is that the creation takes time and effort and requires original thinking, so the copyright holder should have the opportunity to earn money for it. The second is that it is unfair for some other individual or organisation to reproduce the work and to make money from it without any payment to the original creator.

As with the case of data protection discussed in Chapter 9 (Section 9.01), laws are needed to protect copyright. Different countries have different details in their laws but there is an international agreement that copyright laws cannot be avoided, for example by someone publishing the work in another country without the original copyright holder's permission.

Typical copyright laws will include:

- a requirement for registration recording the date of creation of the work
- a defined period when copyright will apply
- a policy to be applied if an individual holding copyright dies
- an agreed method for indicating the copyright, for example the use of the © symbol.

When copyright is in place there will be implications for how the work can be used. The copyright owner can include a statement concerning how the work might be used. For instance, the ACM has the following statement relating to the code of ethics discussed in Section 10.02:

This Code may be published without permission as long as it is not changed in any way and it carries the copyright notice. Copyright © 1999 by the Association for Computing Machinery, Inc. and the Institute for Electrical and Electronics Engineers, Inc.

This is one of several possible variations referring to permissions that are granted when the work has not been sold. If someone has bought a copy of a copyrighted product there is no restriction on copies being made provided that these are solely for the use of the individual. A general regulation relates to books in a library, where a library user can photocopy part of a book.

10.05 Software licensing

Commercial software

In one respect commercial software is no different to any other commercial product. It is created and made available for sale by a company that is aiming to make a profit. There is, however, a significant difference. If you buy a computer you become the owner but if you buy software you do not become the owner. The ownership remains with the vendor. As a buyer you have paid for an end-user licence that allows you to use the software. It is normal that

the software license has to be paid for but there are a number of different options that might be available:

- A fee is paid for each individual copy of the software.
- A company might have the option of buying a site licence which allows a defined number of copies to be running at any one time.
- Special rates might be available for educational use.

A company that normally sells the software licence may sometimes provide a license free of charge. There are two possibilities. **Shareware** is commercial software which is made available on a trial basis for a limited time. It might be the full package available at the time or a limited version of it. A beta test version of new software might be considered to come in the shareware category. **Freeware** might be a limited version of a full package or possibly an earlier version. The difference is that there is no time limit for the licence.

Whatever license is obtained by the user of the software the source code will not be provided and the license will define limitations on the use of the software.

Examples of when using commercial software can be justified include:

- The software is available for immediate use and provides the functionality required
- The software has been created to be used in conjunction with already installed software
- There will be continuous maintenance and support provided
- Taking advantage of a shareware offer might allow suggestions to be made as to how the software could be improved
- Freeware can often offer sufficient functionality to serve a user's limited needs.

Open or free licensing

For open licensing there are two major operations under way. Both are global non-profit organisations. They are very similar in what they provide but there is a difference in their underlying philosophies.

The Open Source Initiative makes **open source software** available. The philosophy here is that the use of open source software will allow collaborative development of software to take place. The software is normally made available free of charge. The source code is provided. The user of the software is free to use it, modify it, copy it or distribute it in accordance with the terms defined by the license.

The Free Software Foundation is so-named because the philosophy is that users should be free to use software in any way they wish. The software is not provided entirely free of charge; there is a small fee to cover distribution costs. The **free software** is still open source. However, there is a special feature of the license which is called 'copyleft'. This is the condition that if the software is modified the source code for the modified version must be made available to other users under the same conditions of usage.

Examples of when using open source software can be justified include:

- The full functionality needed can be provided for at most a nominal cost
- The software could provide the required functionality with just a few modifications to the source code
- A consortium of developers are collaborating in producing a new software suite
- The future development of the software or the continuous provision of the existing software is controlled by the user.

Discussion Point:

How often do you think that open licence software is being used? Should it be used more often?

TASK 10.01

Carry out a search to investigate some of the software available under an open licence.

KEY TERMS

Open source software: software provided with the source code

Free software: software provided with the source code when the user is free to use it as they wish

Shareware: software provided free of charge for a limited period but no source code provided

Freeware: software provided free of charge with no time limit for its use but no source code provided

10.06 Artificial intelligence (AI)

Artificial intelligence (AI) depends on and draws from many other disciplines including: philosophy, psychology, neuroscience, mathematics, linguistics and control engineering. The only definitions of AI that are acceptable are at the same time so generalised that they are not very practical. The following is a typical example:

AI concerns the use of a computer or computer-controlled device to perform tasks normally associated with intelligent behaviour by humans.

We will consider five aspects of intelligent human behaviour and discuss some applications of AI that mimic this human behaviour.

Problem solving

One example is the development of a system that can play chess. This can be considered as displaying artificial intelligence but this is only demonstrated because the rules of chess are limited. A computer with sufficient storage capacity and processing power can investigate so many options for a possible sequence of moves that a human cannot compete.

A second example is the traditional form of expert system that, for example, has been developed to aid medical diagnosis. This is supplied with data and rules from living medical experts. The expert system contains more knowledge than is possible for an individual doctor to have. However, if the expert system is given a new situation that is not covered by the data and rules it has been given, it cannot attempt a new or creative approach – unlike a human.

Linguistics

Voice recognition and voice synthesis techniques are already developed and in use. One example is if you phone a help line where you might be answered by a computer. Provided that you answer questions clearly the computer might be able to identify your needs and pass you on to an appropriate human who can help. However, this is a long way away from the computer itself creating new questions based on your answers and providing the help you need.